



***Earn**₹s*

DSA in C++

Loops

Lecture 08

Summary Notes

By - Raghav Sir

Q. WAP to print all the factors of a given number.

$\Rightarrow$ 'i' is a factor of 'n' if $n \% i == 0$

60 → 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60

$\downarrow$
 i

$\downarrow$
 $\text{sqrt}(60)$

$\downarrow$
 n/i

Q, WAP to tell if no. is prime or composite

Ques: WAP to check if a given number is prime or not.

$i = 2$ to $n-1$ tak loop chala

if($n \% i == 0$) composite

$n \rightarrow 1 \ \& \ n$

$12 \rightarrow 1, 2, 3, 4, 6, 12$

Ques: WAP to check if a given number is prime or not.

5
n

false
flag

```
✓ int n;  
✓ cout<<"Enter a number: ";  
✓ cin>>n;  
✓ bool flag = false; // false means prime  
✓ for(int i=2;i<=n-1;i++){  
    if(n%i == 0){ // factor mil gaya except 1 and n  
        flag = true;  
        break;  
    }  
}  
✓ if(flag==!=true) cout<<"Composite Number";  
else cout<<"Prime Number";
```

i = 2 3 4 5

Output

Enter a number 5

Prime Number

Ques: Print all even numbers from 1 to 100

Trust

Raghav

Class

While Loop (for loop ka replacement)

```
✓ int i = 1;  
  while( i <= 4 ) {  
    |   cout << i;  
    |   i++;  
    |  
  }
```

5
4
3
2
1
i

Output
1 2 3 4

Do-While Loop

आपकी याद नहीं आएगी

Infinite Loop



jo chalta rahe..

Predict the output:

```
int main() {  
    while ( 'a' < 'b' )  
        cout << "malayalam is a palindrome" << endl ;  
}
```

Infinite Loop

Predict the output:

```
int main() {  
  ✓ int i;  
  ✓ while (i = 10) {  
    cout << i << endl;  
    i = i + 1;  
  }  
}
```

Handwritten diagram showing a box containing the value 10, with an arrow pointing to it from the variable *i* in the code above. Above the box is the number 11, and below it is the number 10.

10
10

infinite

Predict the output:

HW

```
int main() {  
    int x = 4, y = 0, z;  
    while ( x >= 0 ) {  
        x-- ;  
        y++ ;  
        if ( x == y )  
            continue ;  
        else  
            cout << x << " " << y << endl ;  
    }  
}
```

Predict the output:

```
int main() {  
    int x = 4, y = 0, z;  
    while ( x >= 0 ) {  
        if ( x == y )  
            break;  
        else  
            cout << x << " " << y << endl;  
            x--;  
            y++;  
    }  
}
```

HW

Predict the output:

```
int main() {  
    int t = 10;  
    while(t/=2){  
        cout << "Hello" << endl;  
    }  
}
```

hw